

IT314 Software Engineering (3-0-2-4)

Autumn 2020

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Course Prerequisite - Basics of Programming, OOP concepts.

The Software Engineering course introduces the basic principles, practices, tools and techniques required to engineer large complex software systems. The course is project-intensive, where students learn by example and practice. The main objective is to understand and learn how complexity and change are engineered during large software development. Here, we would focus on the methodologies (processes), techniques (methods), and tools that can be used to successfully design and validate large software. Wherever relevant, we will make use of various technologies (e.g., DevOps, CASE) to represent various aspects of software development.

The contents to be covered are: (1) Software Requirements Modeling and Specifications, (2) Software Architecture and Design Patterns, Software Development Methodologies, (3) Software Measurement and Metrics, (4) Computer Aided Software Engineering and Tool Support (DevOps, Automation), (5) Software Quality Standards and Quality Assurance, (6) Applications of ML and AI in analyzing software products.

Preamble

Writing programs to fulfil some academic requirements for an academic degree is a radically different exercise compared to how state-of-art software is developed in industry. The main objective of this course to understand and learn how complexity and change is engineered during large software development. Here, we would focus on the methodologies (processes), techniques (methods), and tools that can be used to successfully design and validate large software. Wherever possible, we will make use of UML to represent various aspects of software development.

Marking Scheme:

First-In Semester Exam - 10%; Second-In Semester Exam - 10%; End Semester Exam - 30%; Course Project Work - 40%; Lab, assignments (or presentations), experiments – 10%;

Course Outcomes:

1. The students are expected to learn and practice how the large scale software systems have been developed, what are the related complexities and how to deal with them.
2. The students are expected to learn and understand software development using components.
3. The student is expected to learn and realize the evaluation, analysis, and application of existing technologies and tools in the context of a problem.

Textbooks:

- Shari Lawrence Pfleeger, and Joanne M. Atlee, Software Engineering: Theory and Practice, 3rd Edition, 2006, Pearson.

- Pressman, Roger S. Software engineering: a practitioner's approach. Palgrave Macmillan, 2005.

Course Project Outline:

Specific to the software process model chosen for development of the course project. For example, for Agile Process Model (SCRUM)

1. Requirements in the form of user story (both functional and non-functional)
2. Acceptance Criteria
3. Burn-down chart
4. Daily SCRUM planning and development of sprints.
5. Use of GitHub and DevOps tools.
6. LLMs
7. Slack

Marking Scheme for Projects:

1. The group mark is everybody's mark, i.e., individual assessment of 25% and group assessment of 15%.
2. Everybody reports what they personally did, and separate marks are given to those components by the instructor. This can be done by using either a dossier of rough work or a division of labour report produced by team members.
3. Other group members report (confidentially or openly) the relative contributions of other group members to allow for an adjustment of the final grade.
4. Pop quizzes in class to ensure that students know the intimate details of the project.
5. Cross-validating with the results of individual work (possibly reducing the weight of group work for students who perform poorly on exams or individual assignments).